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Simulating Proppant Flowback for Conventional, Screenless and Multizonal Fracpack Using Novel Resin Coated Proppant

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Abstract

Brown fields with unconsolidated reservoirs often produce high amount of sand and fines. Typical conventional screened Multizone Single Trip (MZST) completion technique is cost-intensive and erodes project economics. Screenless fracpack via frac sleeves offers a solution to significantly reduce well cost while at the same time achieving downhole sand control. Proppant flowback in a screenless fracpack application is a major concern that can curtail production and hence, negatively affects the overall project economics. Mathematical models and laboratory tests to simulate flowback of a novel resin-coated proppant (RCP) are performed to ascertain production boundaries and avoid proppant flowback. The engineering-based proppant flowback studies have enabled the project team to potentially reduce up to 25% of wells cost by performing screenless fracpack via frac sleeves.

Recent literatures on proppant flowback studies have shown that frac width, proppant size, proppant pack permeability, proppant friction coefficient, confining stress and producing fluid viscosity are relevant parameters that determine the critical velocity for proppant flowback. Proppant 40/70 RCP pack is sandwiched between sandpaper, polymer-layered metal plates to mimic the closure stress acting on the RCP pack. Mixture of heated water and glycerol is pumped through the flowback cell with increasing flowrate and measured differential pressure. The free proppant is collected on a fine mesh metal sieve. From the laboratory test, the critical velocity for RCP flowback required for mathematical model and simulations is obtained. The model is then compared with an actual screenless fracpack via frac sleeves job in Field B.

Different levels of curing strength of the RCP are prepared with the "least curing strength" (65 C and 4-hour curing time) whereby the "strongest curing strength" (85 C and 24-hour curing time), which these parameters is based on the parameters of each targeted reservoir zone of each well. The least curing strength's RCP pack shows instability with low critical velocity while the strongest curing strength's RCP pack shows high critical velocity beyond 63 cm/s and high pressure drop at failure. The associated critical velocities are incorporated into the flowback simulator and modelled with specific geomechanical properties of the reservoir. The results show that the longer RCP curing time of beyond 48 hours, with longer drawdown duration of at least 6 hours and maximum drawdown of 300 psi can prevent proppant flowback and ensure near wellbore area is maintained to be propped. The recommendations of the laboratory and simulation

studies were performed in Field B where very limited proppant flowback was observed. The success of the RCP flowback simulation backed by actual field results paves the way for more screenless fracpack application which significantly improves project economics

This is the first RCP laboratory test conducted whereby the results were incorporated into the flowback simulator. Previous studies only involved standard non resin coated ceramic proppants.

Introduction

Proppant flowback is a major concern in wells that are hydraulically fractured whereby these are typically unconventional wells with tight formation. Ensuring the near wellbore area to be fully propped is important due to concerns of tensile failure rock and pinch point in the fracture (Chuprakov et al. 2018). A pinched near wellbore area whether due to excessive over-displacement of proppants during pumping operations or during production due to massive flowback of proppants can lead to a compromised production. Traditional beliefs in unconventional business are that such events are not very prominent. There have been assumptions that once the proppant pack is fully concentrated within the fracture, it can no longer flow. However, there have been field experiences (Potapenko et al. 2017, Potapenko et al. 2019) whereby proppant flowback is supported and even verified by laboratory experiments (Guo and Ugwu 2015) as well as numerical modeling (Parker et al. 1999). These experiments have shown key variables for proppant flowback namely fracture closure stress or the confining stress, proppant adhesion properties, proppant size and the fracture width.

In conventional fracpack wells, the importance of having the near wellbore fully propped is even more paramount. Fracpack wells typically deal with soft and highly permeable formations whereby the Unconfined Compressive Strength (UCS) is less than two- thousand psi. An unpropped near wellbore in a soft formation of sandstone pinches out almost immediately once the pump is turned off. In a conventional fracpack operation, the fracture width is significantly higher than in an unconventional or tight formation. Hence, the tensile stress acting on fracture of the unpropped near wellbore is higher. The higher the fracture confining stress exerting on the proppants, the higher the critical velocity (Zhang and Guo 2016). This appears to be intuitive, however, it is important to take note that other experiments show the opposite to be true (McLennan et al. 2015) whereby as confining stress increases, the critical velocity decreases. Milton-Tayler 1992 has summarized that the sensitivity of the confining stress only applies at around 500 to 2,000 psi. For unconventional rocks where the UCS and confining stress typically goes to 20,000 psi, the effect of confining stress on the critical velocity no longer applies.

In this paper, there are four steps to simulate proppant flowback using a novel RCP for Well B, offshore Peninsula Malaysia. The first step is simulating the fracture geometry and proppant placement within the fracture using a multiphysics hydraulic fracturing software. The second step is performing a laboratory experiment using the novel RCP whereby the results of the experiments are used in the proppant flowback simulator. The third step is simulating the proppant flowback using a rigorous proppant flowback model (Chuprakov et al. 2020). The final step is comparing the results of the proppant flowback simulator and the results of the actual production for Well B.

Well B is the pilot well where smart fracturing sleeves are utilized to perform screenless fracpack. The challenge here is to ensure that the slurry is under-displaced in such not to leave much proppant in the wellbore that requires massive proppant cleanup operations via coiled tubing. However, as discussed at length above, care must be taken not to over-displace the slurry leaving the near wellbore unpropped which could jeopardize productivity. Typically, the field where Well B is drilled has wells which are completed with MZST screened completions with Fracpack. The technique though effective but is very cost intensive. As the field becomes marginal in reserves, an alternative and a more unconventional technique is sought after which significantly reduces the well cost and subsequently helps improve project economics.

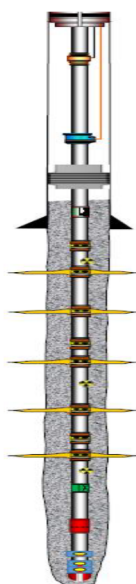


Figure 1—Well B Completion Schematic

Methodology and Results

Fracturing Model Simulation

Geomechanics and petrophysics studies were conducted on this field, where the properties of the target zones obtained are used as inputs of the fracture modelling. Sensitivity of reservoir properties effect on fracture geometry was modeled. The fracturing simulation uses multiphysics modelling to predict the proppant distribution inside the fractures. Results are presented below

Table 1 shows the results of fracture geometry simulation for three zones whereby each zone is simulated with softest and hardest possible rock properties.

Table 1—Fracture Geometry.

Interval	Fracture width realisation (formation softness in bracket)	Max Hyd Xf	Prop Xf	EOJ Hyd H at well	EOJ Hyd W at well	Prop W at well	Avrg Prop W	EOJ Pnet	Eff Fluid
		m	m	m	m	mm	mm	atm	
Upper (I-100)	Thickest (Softest)	24.7	24.7	9	56.1	27.3	27	133	21.4
	Thinnest (Hardest)	29.6	29.6	14.4	29	14.2	12.9	104	20.8
Mid (I-105)	Thickest (Softest)	18.1	18.1	7.7	89	31.8	29.6	250	33.5
	Thinnest (Hardest)	20.2	20.2	19	36.1	11.7	11	95	34.6
Bot (I-122)	Thickest (Softest)	21.2	21.2	16.6	44.7	14.3	17.7	67	24.6
	Thinnest (Hardest)	18	18	17.7	26.7	15.1	16.2	79	18.6

As shown on Figure 2 above, the proppant distribution profile inside the fracture is obtained for each softest and hardest case of all the zones. Once the fracture geometry and proppant distribution within the fracture geometry are determined, the next step is to conduct laboratory experiments to obtain specific constants for the tested RCP. These constants, together with the fracture geometry and proppant distribution properties data obtained from the simulation are then plugged into the proppant flowback simulator. The flowback simulator simulates different sensitivity values for both drawdown pressure and drawdown period. Moveable proppant mass for all cases is then determined based on the results of the proppant flowback simulator.

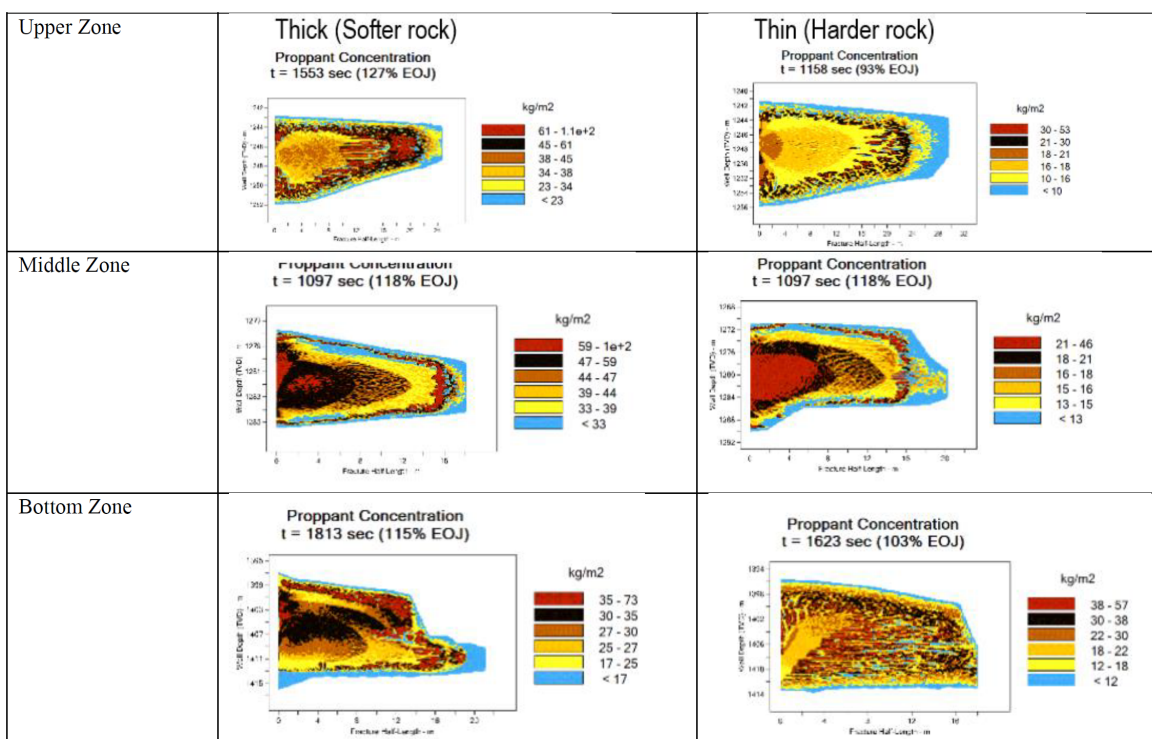


Figure 2—Proppant concentration in both scenarios of softer and harder formation

Laboratory Experiments

The laboratory experiment is aimed at replicating the RCP curing strength and well startup conditions with proppants under fracture closure stress and drag from unloading flowing liquid. The objective is to obtain the critical velocity and calibration factors for four different RCP groups based on their curing strength namely the "very low", "low to medium", "medium" and "strongest" states. The RCP with different curing strengths is also subjected to a simulation on the speed of the well start up whether "gradual" or "fast". The expectation is a "gradual" start-up of well unloading should produce less proppant flowback compared to "fast" start-up as it allows time for the proppant pack to stabilize.

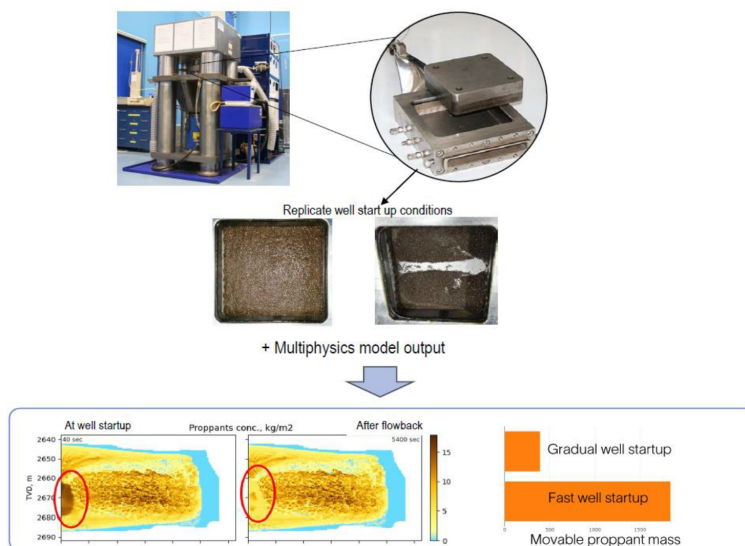


Figure 3—Laboratory objectives and results

The laboratory equipment setup can be seen in Figure 4. The RCP is placed inside the flowback cell where then it is subjected to certain flow conditions to determine flowback. In a nutshell, two plates sandwich a proppant pack in between where closure stress is applied to the proppant pack. The closure stress is kept stable by an electronically controlled syringe pump through the duration of the tests. A heated mixture of tap water and glycerol is pumped through the flowback cell by a water pump. Both pumping temperature and pressure are recorded at the inlet face of the proppant pack to enable differential pressure calculation across the flowback cell. Free proppants are collected and weighed on a fine mesh metal sieve. To mimic both "softer" and "harder" rocks, a sandpaper was glued both ends of the plates to imitate the "softer" rock while bare steel plates are used to imitate the "harder" rock. Methods and equipment setup are described in detail by Chuprakov et.al 2020.

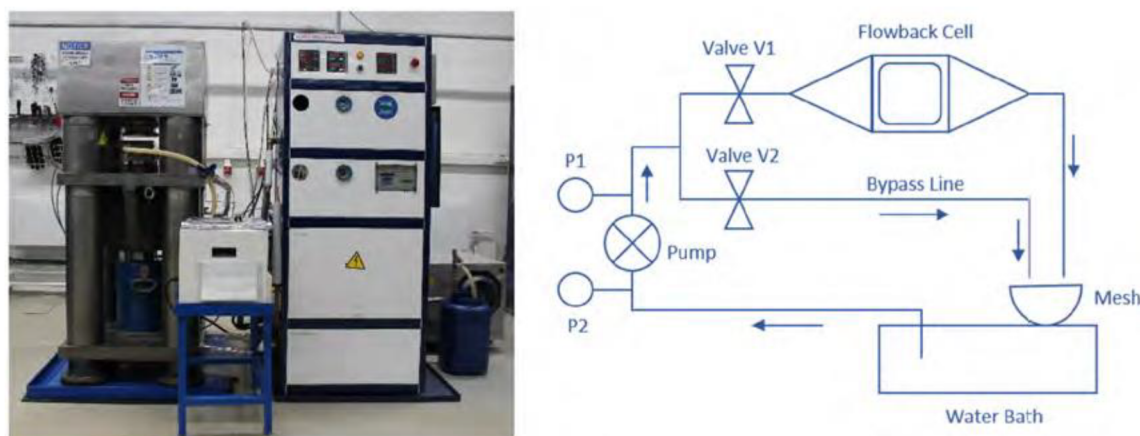


Figure 4—Laboratory equipment set up (Chuprakov et al. 2020)

Table 2 shows that RCP test condition is separated into four groups. All groups are subjected to the same applied stress of 1,000 psi. However, the time where the RCP is subjected to stress differs from 4 hours to 24 hours. The longer curing time implies that the curing strength is the strongest while the shortest curing time implies lower curing strength. Between "very low" and "low to medium" curing strength groups, the temperature is increased higher up to 85 C. For the "low to medium" group, the higher temperature implies stronger curing strength. The results will compare the performance of all four curing strength groups from "very low" to the "strongest" case. Note that, all test parameters are selected to simulate the condition each top – middle – bottom target reservoir.

Table 2—RCP separated by four groups based on curing strength expected

Curing Strength expected	w/d	Applied Stress, psi	Temperature, °C	Time, hours
Very Low	10.5	1000	RT → 45 → 65	4
Low to Medium	11.0	1000	RT → 45 → 65 → 85	4
Medium	10.9	1000	RT → 45 → 65 → 85	6
Strongest	10.5	1000	RT → 45 → 65 → 85	24

Table 3—Non-RCPs properties from previous experiments (Chuprakov et al. 2020)

Proppant type	Light-weight proppant (LWP)	Light-weight proppant (LWP)	Intermediate strength proppant (ISP)	Fracturing sand
Mesh size	20/40	12/18	12/18	20/40
Average diameter, in.	0.029	0.055	0.053	0.022
Absolute density, g/cm ³	2.70	2.70	3.23	2.65
Bulk density, g/cm ³	1.55	1.63	1.92	1.58
Roundness	0.80	0.90	0.89	0.91
Sphericity	0.90	0.90	0.87	0.82

Figure 5 (Chuprakov et al. 2020) shows previous experiments of conventional proppants (non-RCP) where typically when the ratio of fracture width, w over proppant diameter, d (w/d) at around 2.5 to 3, the critical velocity increases exponentially. It suggests that at w/d ratio around 3, the proppant flowback is almost impossible to happen due to massive liquid velocity required to dislodge the proppant.

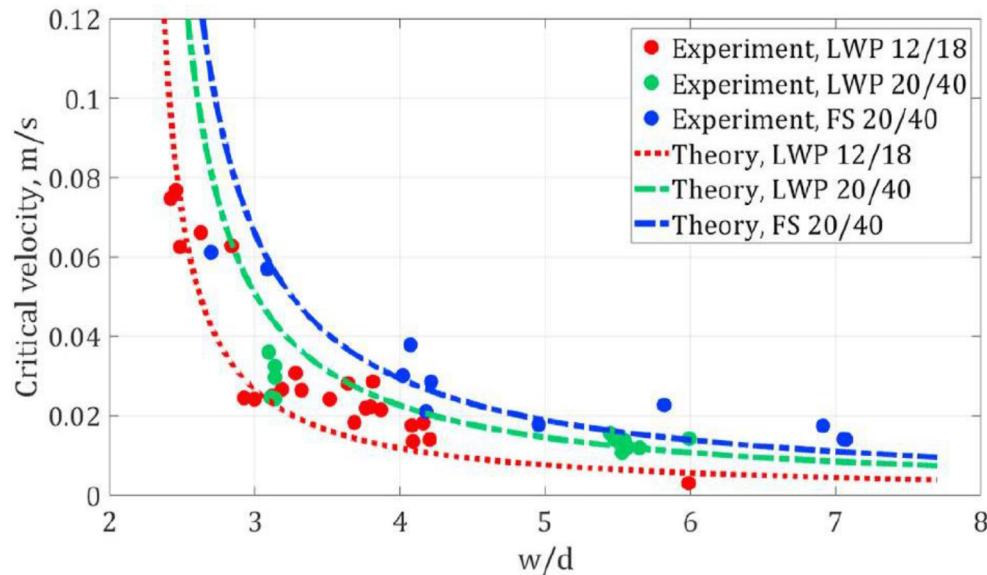


Figure 5—The effects of fracture width over proppant diameter ratio on critical velocity (non-resin coated proppants)

Since the RCP is tested for conventional fracpack application with relatively soft and permeable formation, the fracture width is expected to be much higher than a fracture width for an unconventional formation. Hence as Table 1 shows, the w/d ratio used is approximately around 11 whereby the critical velocity should be around 0.01 m/s or 1 cm/s.

Based on Figure 6 above, for case of "very low" curing strength group, the RCP proppant pack failure occurs at 2 psi (0.15 kgf/cm²) at a fluid velocity rate of 5.4 cm/s. As mentioned earlier, the conventional non-RCPs at w/d of 11 should fail at 1 cm/s but here the RCP shows a superior performance as it fails at a higher fluid velocity. The result is expected as RCP has higher cohesion properties and that should increase the critical velocity of the RCP compared to non-RCPs. Figure 7 shows a channel of RCP pack being created indicating that the RCP pack fails to be retained and flowback of RCP occurs at the one end of the tray

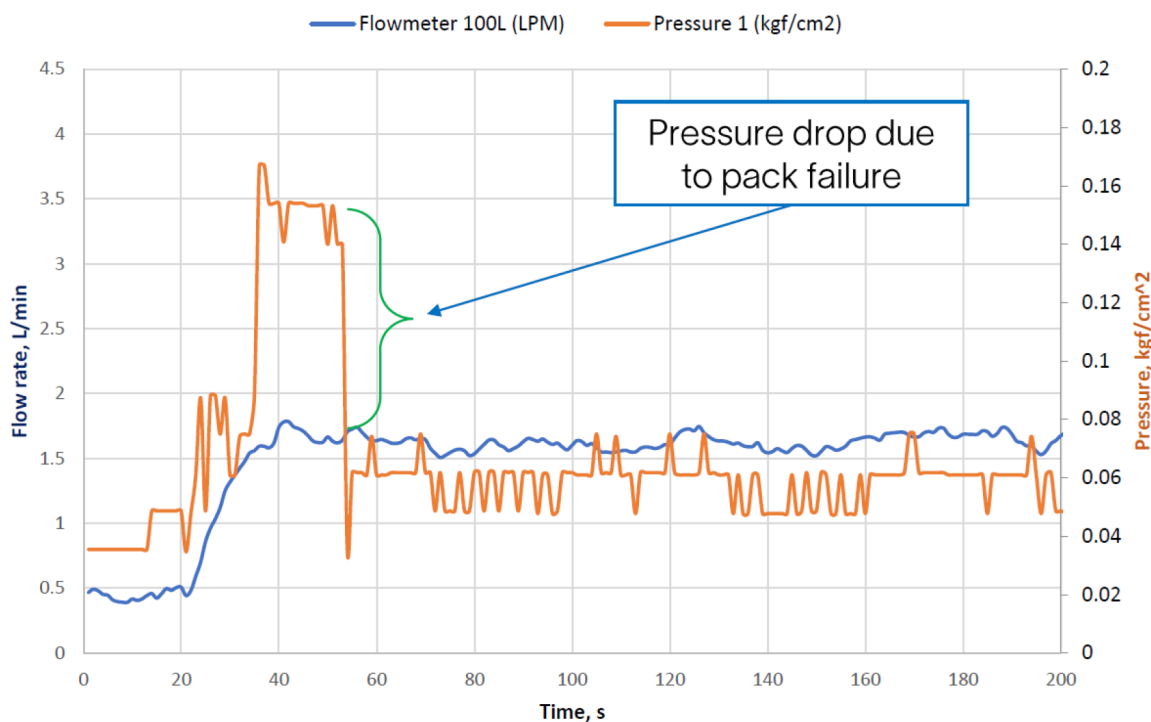


Figure 6—Flowback test results for the "very low" curing strength group of RCP

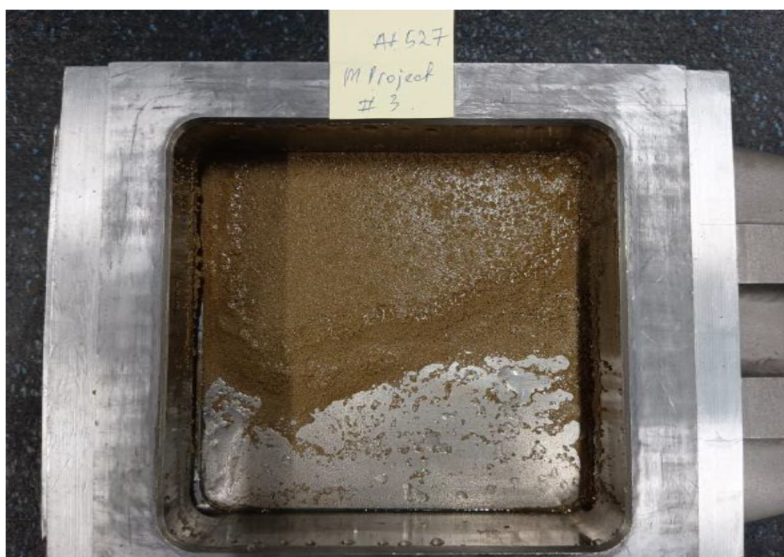


Figure 7—Channel of RCP flowback for the "very low" curing strength group

Figure 8 shows the "strongest" RCP flowback test whereby the critical velocity has not yet been achieved although the flow velocity has reached 63.8 cm/s. If this value is conservatively assumed as the critical velocity for the "strongest" RCP group, the corresponding pressure at such velocity is at 197 psi. Figure 8 shows that the RCP pack within the tray remain intact and RCP flowback does not occur. The test results from both "very low" and "strongest" RCP curing strength are used as lower and upper bounds of critical velocity of the proppant flowback modelling parameters.

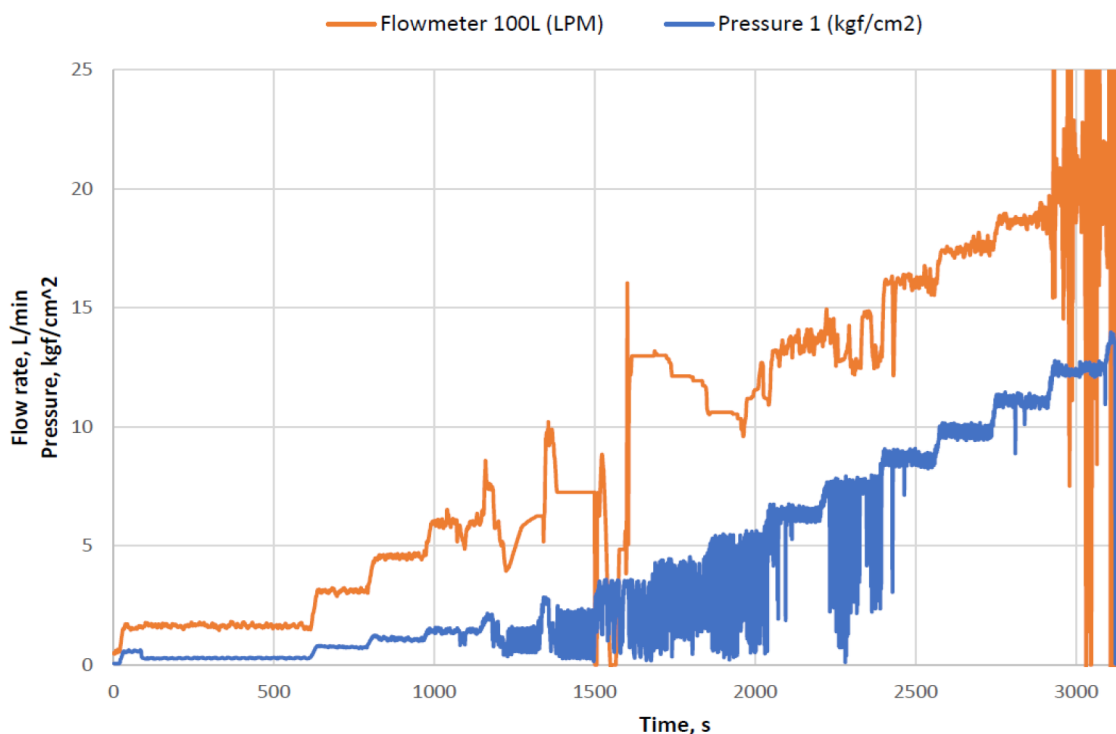


Figure 8—Flowback test results for the "strongest" curing strength group of RCP

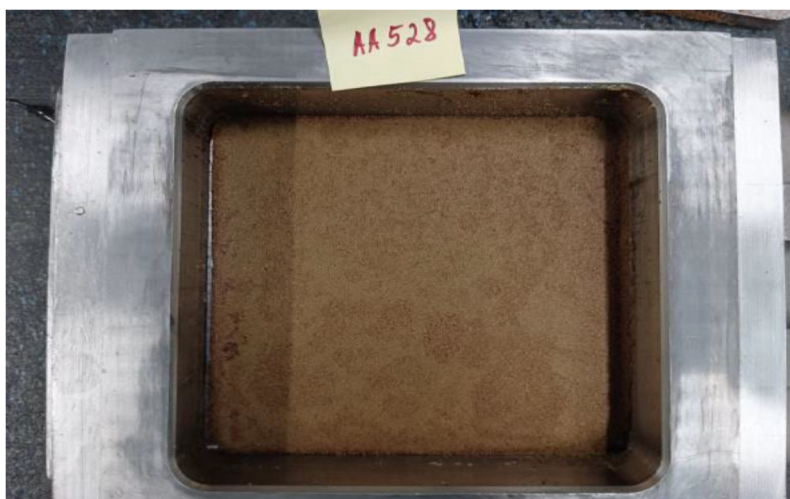


Figure 9—RCP pack remains intact for the "strongest" curing strength group

Flowback Modelling

The flowback modelling is focused and sensitized mainly on the three perforation intervals (upper, middle and bottom), two formation properties (softest and hardest) as well as two curing scenarios (cured RCP and non-cured RCP). The sensitivity analyses were also performed based on the drawdown pressure (20 bar and 40 bar) and drawdown duration (1 hour, 3 hours and 5 hours). Figure 10 and Figure 11 below illustrate the two opposite conditions of the proppant flowback, one with the non-cured RCP and the other with cured RCP

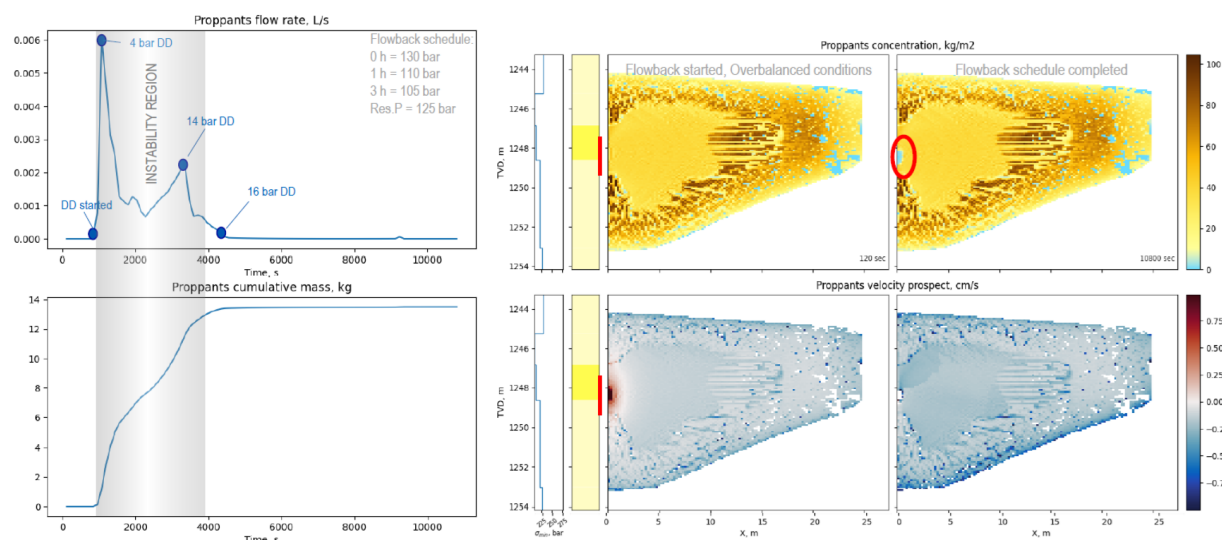


Figure 10—Flowback modelling – Upper interval, soft rock and non-cured RCP case

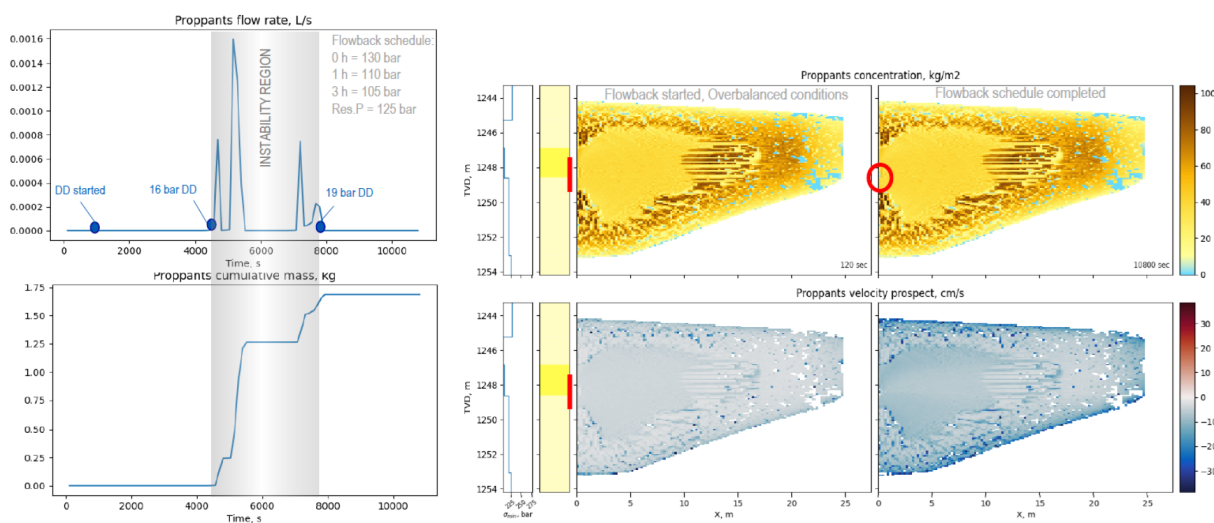


Figure 11—Flowback modelling – Upper interval, soft rock and cured RCP case

The upper interval (Figure 10), soft rock and non-cured RCP case only has a critical velocity of 0.27 cm/s. It shows that proppant flowback happens after 1000 seconds (27 minutes) of the well unloading at 4 bar (58 psi) drawdown pressure near the wellbore area. The area most affected by the flowback is indicated by the red circle. This is the region where the closure stress subjected in the near wellbore is the weakest and where velocity and drag are the highest.

The upper interval (Figure 11), soft rock and cured RCP has a critical velocity of 7 cm/s. It shows that proppant flowback happens after 4500 seconds (1 hour, 15 minutes) of the well unloading at 16 bar (232 psi) drawdown pressure. A much longer period and very minimal affected area compared to the non-cured RCP case. Similar simulation iterations were then applied to the middle and top zones, where the results are shown in Table 4 below:

Table 4—Sensitivity analyses and results of flowback modelling

Scenarios					Results		
Interval	Formation option (fracture width)	RCP curing scenario	Drawdown during flowback, bar	Drawdown increase duration, hours	Drawdown ranges with mobilization spikes, bar	Movable proppant, in kg	Dangerous zone extension (along X x along Z), m
I-100 (upper)	Soft (thick)	Non-Cured	20	3	2 to 14	30	1 x 2
		Non-Cured	40	5	2-14, 22-35	35	1.5 x 2.5
		Non-Cured	20	1	2 to 18	40	1.5 x 2.5
		Non-Cured	40	1	2 to 25	45	2 x 3
		No RCP effect	20	3	1 to 15	60	7 x 8
		No RCP effect	20	5	1 to 15	70	7 x 8
		No RCP effect	40	5	1-15, 20-40	70	15 x 9
		Cured	20	3	16 to 19	7	0.3 x 0.3
	Hard (thin)	Non-Cured	20	3	2 to 8	20	1 x 1.5
		Cured	20	3	16 to 19	6	0.3 x 0.3
I-122 (bottom)	Soft (thick)	Non-Cured	20	3	2 to 10	15	1 x 4
		Cured	20	3	15 to 17	2	0.2 x 0.3
	Hard (thin)	Non-Cured	20	3	1 to 8	10	2 x 2
		Cured	20	3	Beyond 20	1	0.2 x 0.2
I-122 (bottom)	Soft (thick)	Non-Cured	20	3	2 to 12	15	5 x 3
		Cured	20	3	>19	1	0.2 x 0.3
	Hard (thin)	Non-Cured	20	3	2 to 11	12	2 x 2
		Cured	20	3	Beyond 20	1	0.2 x 0.2

Safe option, ensure 48 h curing time
Safe option, not less than 60 h curing time is preferable
Unsafe option, impact on production
Unsafe option, impact on production, risk of uncontrolled proppant flowback
Unsafe option, impact on production, high probability of uncontrolled proppant flowback

The simulation iterations are conducted to reflect the condition of RCP curing strength, drawdown pressure and duration for each of the target sand. An operating envelope, from safe to unsafe option, for each of the zone are also determined. The summary shows that the upper zone is the riskiest zone which requires longer curing time of at least 60 hours to avoid massive proppant flowback. A prolonged proppant flowback will negatively affect the production from the zone. From the simulation results, it could be concluded that an aggressive drawdown during flowback should also be avoided to minimize proppant flowback. Proper and calculated bean up procedure is important to avoid unwanted proppant flowback.

Actual Results

After the fracpack treatment executed, well B was successfully unloaded and has been producing since December 2023. The well was unloaded after 96 hours (4 days) of well completion to ensure the RCP packs within the fractures were fully cured. The drawdown pressure was kept at around 250 psi, well below 20 bar (290 psi) to ensure that the upper zone is within its safe envelope. The well has been producing sand-free and proppant-free until today.

Conclusion

The screenless frac sleeves completions for conventional fracpack application is made possible because of the rigorous design and engineering on proppant flowback. Laboratory experiments on novel RCP was performed to obtain the lower and upper limit of the proppant flowback based on the RCP curing strength. Multiphysics fracture modelling is used to predict the proppant distribution inside the fracture. Data from both lab test and fracturing simulator is then used as inputs to the proppant flowback simulator to ascertain safe envelopes of production for each of the target zone. Finally, ensuring all the simulation result and operating envelope being implemented on the actual well operation plays a critical role for the success of the job. More types of RCPs should be tested and simulated in the future for comparison.

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